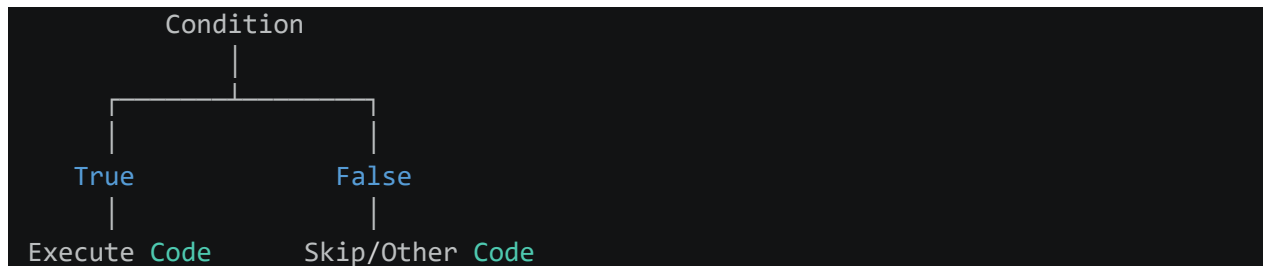


❖ Decision Control Instructions

- Decision Control Instructions help a program decide which code should be executed based on a condition.



1. if Statement

- The if statement executes a block of code only when the condition is True.
- Syntax
if condition:
 statement

```
age = 20  
  
if age >= 18:  
    print("Eligible for Voting")
```

2. if-else Statement

- The if-else statement executes one block if the condition is True, otherwise it executes another block.
- Syntax
if condition:
 statements
else:
 statements

```
age = 16  
  
if age >= 18:  
    print("Eligible")  
else:  
    print("Not Eligible")
```

3. if-elif-else Statement

- Used when there are **multiple conditions**.
- Python checks conditions one by one.
- Ex

```
marks = 72

if marks >= 90:
    print("Grade A")

elif marks >= 75:
    print("Grade B")

elif marks >= 35:
    print("Grade C")

else:
    print("Fail")
```

4. Nested if Statement

- A Nested if means writing one if statement inside another if statement.
- Ex

```
age = 20
citizen = True

if age >= 18:

    if citizen:
        print("Eligible for Voting")
```

5. Match Case

- match-case is used as an alternative to writing multiple if-elif-else statements when checking the value of a single variable.

```
day = 3

match day:

    case 1:
        print("Monday")
```

```
case 2:  
    print("Tuesday")  
  
case 3:  
    print("Wednesday")  
  
case _:  
    print("Invalid Day")
```

6. Conditional Expression (Ternary Operator)

- A **Ternary Operator** is a short way of writing an if-else statement in a single line.
- Ex

```
age = 20  
  
result = "Eligible" if age >= 18 else "Not Eligible"  
  
print(result)
```